

ML Scoring for Proposal Development in the Consulting Industry

AI/ML Business Use Case: Part Two

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Preamble

The **Part One** of this activity has been attached in the [Annex](#) section of this paper. It covers the solution proposal to automate opportunities identification in the consulting industry, and score them for the business development unit.

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Submitted to: Professor Jes Black (Machine Learning & AI)

1. Executive Summary

This paper presents a machine learning solution designed to enhance opportunity identification within a management and economics consulting firm. Recognizing the inefficiencies of manual RFP scoping and prioritization, the project proposes an AI-driven system that automatically scores incoming proposals based on their likelihood of success, enabling business development teams to allocate resources more strategically.

Two classification models were trained—logistic regression and random forest—with the latter significantly outperforming the baseline in accuracy, recall, and ROC-AUC. The random forest model demonstrated robust predictive performance and highlighted key drivers of success, including contract size, region, and contract type.

Deployment is proposed through a cloud-based platform integrated with existing BD workflows, supported by user training, a feedback loop, and governance measures. The model is expected to save over \$750,000 annually in staff hours while improving win rates and decision-making efficiency. Future recommendations include incorporating advanced NLP techniques, expanding data sources, and exploring adjacent use cases to deepen the firm's data-driven capabilities.

2. Data Preparation

A suitable dataset was required to closely mimic a real-world repository of procurement opportunities to implement the proposed machine learning solution for improving opportunity identification in a consulting firm. For this purpose, publicly available procurement data from the World Bank was used as a foundation. Specifically, the dataset selected was the [Corporate Procurement Contract Awards](#) data, which contains detailed records of contracts awarded through World Bank procurement processes. While the original dataset only included awarded contracts, it was repurposed as **Company A**'s internal dataset of all proposals submitted, including successful and unsuccessful bids. Company A is our case study consulting firm for this activity.

The initial dataset contained fields such as award date, contract description, contract amount, funding source, commodity category, and organizational unit within the World Bank. These fields were particularly useful for modeling, as they captured both financial and contextual attributes of procurement opportunities. Several columns were removed from the original dataset to reflect the point of view of a

single consulting firm, including information such as supplier names and countries, as these would always be assumed to refer to the hypothetical company submitting proposals.

[5]: # Load raw business case dataset
df = pd.read_csv("business_case_dataset_CSV.csv")
df.head()

	Award Date	Commodity Category	Contract Description	WBG Organization	Contract Award Amount (USD)	Fund Source	VPU description
0	07/08/2021	CONSULTING	Angola Telecom Strategic Options Study	IBRD	311624.0	WBG	Infrastructure VPU
1	07/13/2021	CONTRACT CONSULTANTS	IFC Digilab Finance	IFC	500000.0	WBG	LAC & EUR
2	07/13/2021	CONTRACT CONSULTANTS	IFC Digilab Finance	IFC	500000.0	WBG	LAC & EUR
3	07/21/2021	CONSULTING	Kenya Electricity Demand Stimulation for Last ...	IBRD	293825.0	Trust Fund	Off. of Regional Vice President
4	07/21/2021	CONSULTING	Critical Information Infrastructure Protection...	IBRD	299002.0	Trust Fund	Infrastructure VPU

Figure 1: Raw dataset (without few non-informative fields dropped)

A target variable: **[Proposal Status]**, was introduced to the dataset. And since the original dataset contained only awarded contracts, a synthetic process was applied to create a balanced dataset. A portion of the awarded contracts was duplicated and slightly modified to simulate lost opportunities. The proposal status column labeled These synthetic examples as " Lost, " while the original entries retained the label “Won.”

[33]: df["Proposal Status"] = "Won"

[35]: # Creating synthetic lost proposals

import numpy as np

Sample 50% of the data for synthetic lost cases
lost_cases = df.sample(frac=0.5, random_state=42).copy()
lost_cases["Proposal Status"] = "Lost"

Modify some fields
lost_cases["Contract Award Amount (USD)"] *= np.random.uniform(0.8, 1.2, size=len(lost_cases))
lost_cases["Award Date"] = pd.to_datetime(lost_cases["Award Date"]) + pd.to_timedelta(np.random.randint(5, 60, size=len(lost_cases)), unit="D")

Combine
df_full = pd.concat([df, lost_cases], ignore_index=True)
df_full.head(10)

Figure 2: Generating the synthetic ‘Lost’ proposal status records

Additionally, a geographical context was engineered from the contract description column by extracting country names using the PyCountry library and predefined country lists from the [United Nations](#). Countries were then mapped to broader geographic regions and subregions, enabling the model to incorporate spatial patterns into its predictions. However, not all descriptions referenced a clear country or region. In such cases, the country was labeled as “Unknown.” To address this, records with unknown country entries were dropped, while a new binary column

titled “**Has Known Country**” was introduced to indicate whether the corresponding region was identifiable or not.

Finally, the cleaned dataset was saved for further processing in the machine learning pipeline. This version included the newly created binary outcome variable, which would be used to train and evaluate predictive models. This approach ensured that the dataset aligned with the use case of a consulting firm attempting to prioritize incoming RFPs based on historical success indicators.

3. Exploratory Data Analysis (EDA)

Following the preparation of the dataset, an extensive exploratory data analysis (EDA) was conducted to understand the distribution of proposals, identify potential patterns, and assess the relationships between key variables and the likelihood of a proposal being awarded. The final dataset comprised 3,666 records, split between proposals labeled as “Won” (representing actual contract awards) and synthetic “Lost” proposals generated to simulate unsuccessful submissions. All entries in the dataset contained structured metadata relevant to the procurement context.

[73]:

df = rfp_df
df.head(10)

[73]:

	Award Date	Commodity Category	Contract Description	Organization	WBG	Contract Award Amount (USD)	Fund Source	VPU description	Country	Region	Subregion	Proposal Status	Year	Has Known Country
0	2015-08-26	CONSULTING	Afghanistan Multi-hazard Risk Assessment, Cost...		IBRD	863674.48	Trust Fund	GP- CCSA, Vice Presidency	Afghanistan	Asia	Southern Asia	Won	2015	1
1	2015-09-30	CONSULTING	Afghanistan Support in the Development of a Pu...		IBRD	411805.00	Trust Fund	GP- CCSA, Vice Presidency	Afghanistan	Asia	Southern Asia	Won	2015	1
2	2017-07-25	CONTRACT CONSULTANTS	Afghanistan Consultancy Services for Household...		IBRD	423820.00	TRUST FUND	VP, SD Practice Group	Afghanistan	Asia	Southern Asia	Won	2017	1
3	2018-08-12	CONTRACT CONSULTANTS	Afghanistan Technical Support to Executive Com...		IBRD	349159.00	Trust Fund	VP, EFI Practice Group	Afghanistan	Asia	Southern Asia	Won	2018	1
4	2016-10-07	CONSULTING	Afghanistan Service delivery and classroom obs...		IBRD	310771.00	WBG	VP, HD Practice Group	Afghanistan	Asia	Southern Asia	Won	2016	1

Figure 3: Snapshot of the dataset after data preparation, before EDA

One of the first steps in the EDA was to assess the distribution of proposal statuses. While the dataset had originally been imbalanced due to the presence of only successful contracts, the synthetic balancing introduced a realistic scenario in which not all proposals are awarded. This binary outcome formed the target variable for the machine learning models. The data was then analyzed over time to determine

whether temporal patterns existed in the awarding of contracts. By aggregating proposals by year, a time-series trend was observed, with increased proposal activity in more recent years, likely reflecting improved data availability and reporting practices.

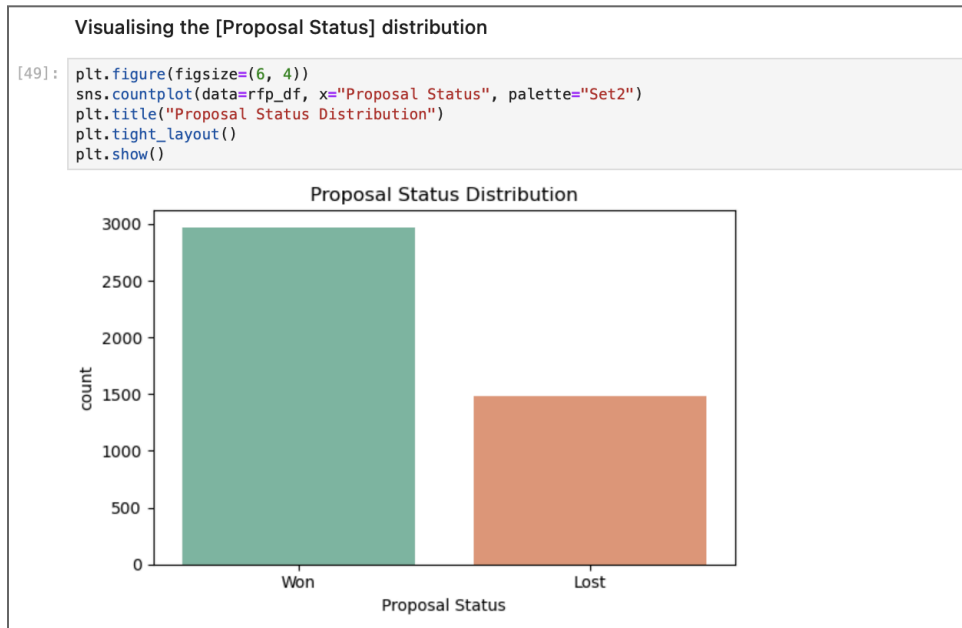


Figure 4: Distribution of proposal status

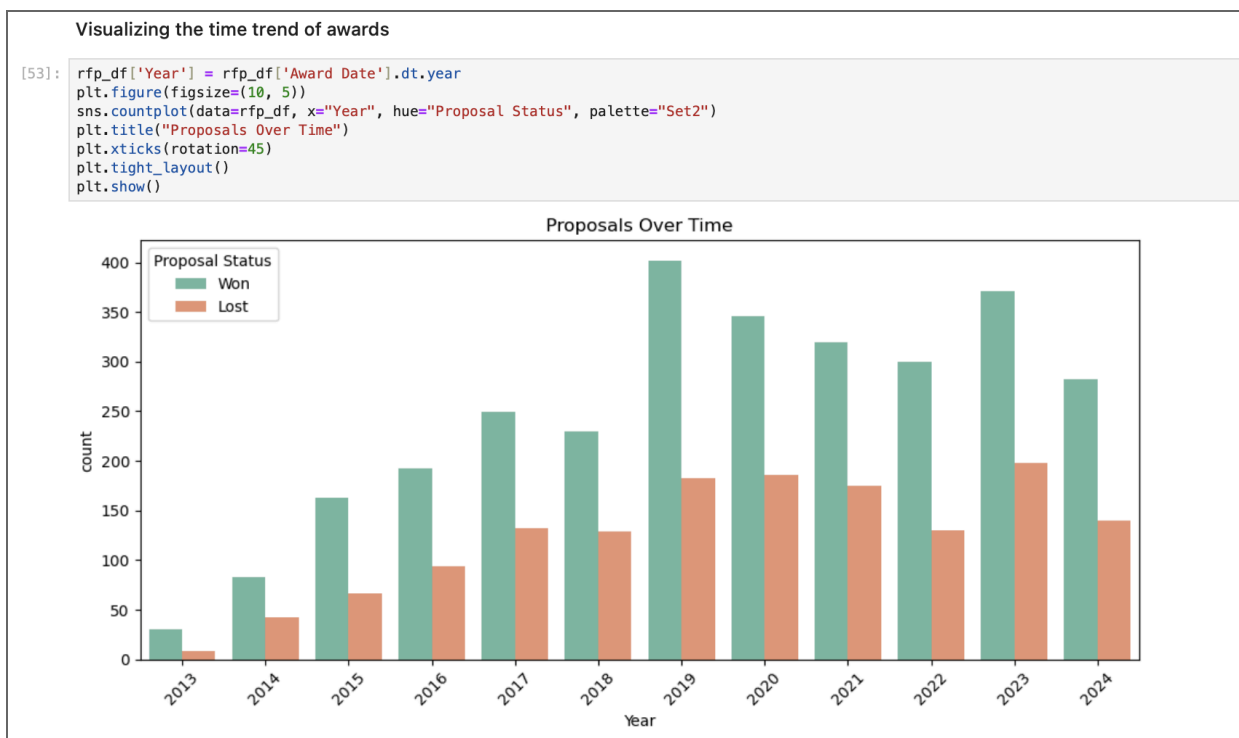


Figure 5: Proposal status over time

Aggregating proposal outcomes across countries and regions also explored geographical distribution. The dataset represented regions such as Africa, Asia, and Latin America well, reflecting the World Bank's global operations. The introduction of the Has Known Country feature allowed the analysis to include a binary comparison between proposals with clearly identified geographic context and those without. Preliminary insights suggested that proposals associated with specific countries or regions were more likely to be successful, indicating the potential influence of localized expertise or previous engagement history.

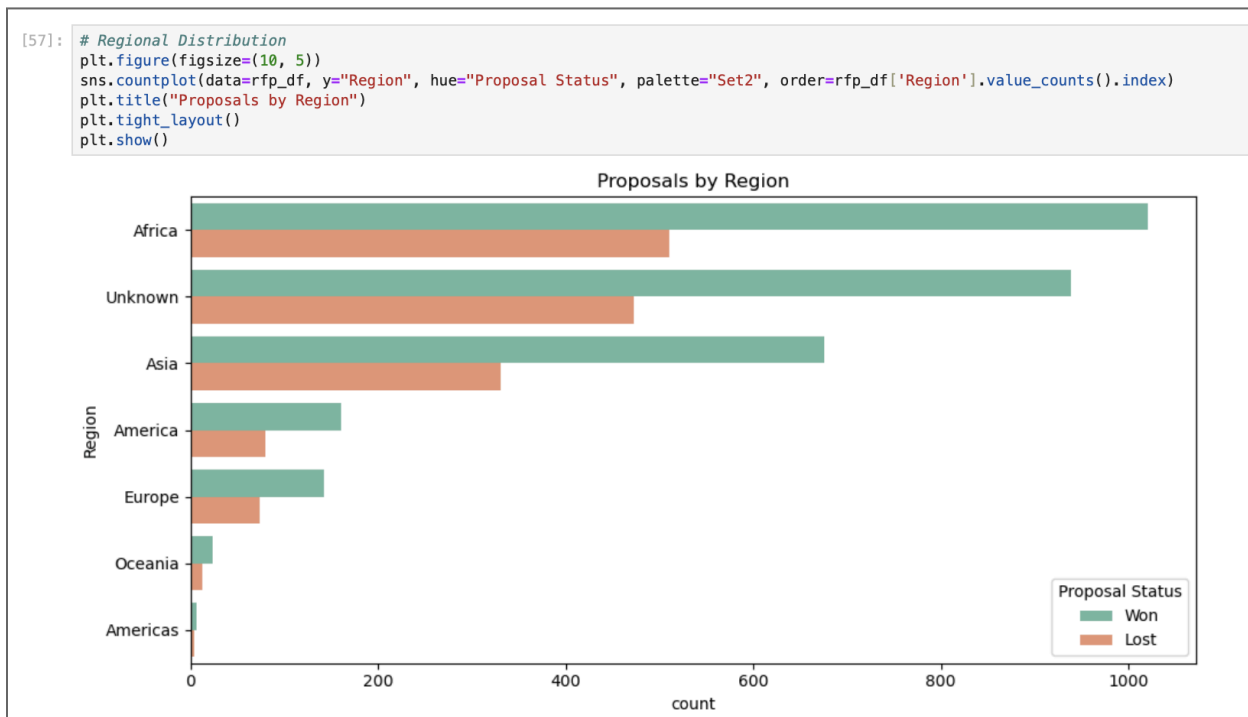


Figure 6: Proposal status by region

Financial attributes of the proposals were also explored through statistical summaries and visualizations. It was observed that the distribution of contract values was right-skewed, with a small number of high-value contracts and a majority of proposals falling in the lower-to-mid financial range. When comparing “Won” versus “Lost” proposals, “Won” proposals exhibited slightly higher median contract values, suggesting a potential correlation between project size and award likelihood.

Overall, the exploratory analysis was a foundational step to inform feature selection, validate assumptions, and guide the modeling approach. The patterns observed in this phase suggested that several features - including contract value, region, and

category, could have predictive power in determining the success likelihood of a proposal. These insights laid the groundwork for developing classification models.

4. ML Model Development

This project phase was building machine learning models capable of predicting the likelihood of a proposal being awarded based on historical procurement data. The primary objective of this modeling exercise was to equip a hypothetical consulting firm with an intelligent system that can evaluate incoming Requests for Proposals (RFPs) and estimate their probability of success, thereby enhancing the firm's ability to prioritize and allocate resources strategically.

The task was formulated as a binary classification problem, where the target variable was the proposal status; either "Won" or "Lost." To train and evaluate models, the dataset was split into a training set (80%) and a test set (20%), ensuring that the class distribution remained stratified to reflect the balanced structure of the target variable.

Two models were selected for development: **logistic regression** and **random forest**. Logistic regression served as a baseline due to its simplicity, interpretability, and ability to reveal linear relationships between features and the probability of a proposal being awarded. The random forest classifier, an ensemble learning method based on decision trees, was chosen for its robustness in handling nonlinear relationships, capacity to model interactions between variables, and effectiveness in working with mixed-type features.

The features used for model training included both categorical and numerical variables. The categorical variables: commodity category, WBG organization, fund source, and region were transformed using one-hot encoding after imputing missing values. Numerical features, including the contract award amount and the binary indicator Has Known Country, were standardized through mean imputation. These transformations were integrated into a preprocessing pipeline using scikit-learn's ColumnTransformer, ensuring that the same preprocessing steps would be applied consistently during training and inference.

Each model was wrapped in a scikit-learn pipeline that combined preprocessing and model fitting. The logistic regression model was trained with default parameters and a maximum iteration threshold increased to ensure convergence. In contrast, the

random forest model was trained with 100 estimators and a fixed random seed for reproducibility.

Following training, predictions were generated on the test set, and model outputs were compared using both class predictions and probability scores. This dual-output approach enabled the subsequent evaluation of classification accuracy and the models' ability to rank proposals based on their likelihood of being awarded. The use of pipelines streamlined the training process and facilitated seamless preprocessing integration, ensuring no data leakage occurred between training and evaluation phases.

Through this development process, both models were successfully trained and prepared for comparative performance evaluation using accuracy, precision, recall, F1-score, and ROC-AUC metrics. These evaluation results are discussed in the next section.

5. Model Evaluation

The performance of the two trained classification models; logistic regression and random forest was evaluated using a variety of metrics including accuracy, precision, recall, F1-score, and area under the Receiver Operating Characteristic curve (ROC-AUC). These metrics were chosen to comprehensively assess overall classification accuracy and the models' ability to differentiate between successful and unsuccessful proposals.

The logistic regression model served as the baseline and achieved an overall accuracy of approximately **66.8%**. While this model demonstrated high recall for the "Won" class, it failed to correctly identify any of the "Lost" proposals in the test set. The precision for the "Lost" class was effectively zero, indicating that the model predicted every proposal as likely to be awarded. Consequently, the ROC-AUC score was only marginally better than random guessing at 0.51. This result highlighted a critical limitation: logistic regression, in its basic form, was not able to learn sufficient distinction between winning and losing proposals based on the selected features.

Logistic Regression Report:				
	precision	recall	f1-score	support
0	0.00	0.00	0.00	244
1	0.67	1.00	0.80	490
accuracy			0.67	734
macro avg	0.33	0.50	0.40	734
weighted avg	0.45	0.67	0.53	734
ROC-AUC: 0.5092422214787554				

Figure 7: Logistic regression model evaluation report

In contrast, the random forest classifier significantly outperformed the logistic model across all metrics. It achieved an overall accuracy of **68.9%**, with a much more balanced classification performance. For the “Lost” class, the model reached a precision of **54.4%** and a recall of **40.2%**. For the “Won” class, precision and recall were **73.6%** and **83.3%** respectively. The ROC-AUC score of 0.64 further confirmed the model’s ability to rank proposals effectively by their likelihood of being awarded. These results suggest that the random forest model is far more suited for deployment in a decision-support context, where accurate identification of both successful and unsuccessful proposals is crucial.

Random Forest Report:				
	precision	recall	f1-score	support
0	0.54	0.40	0.46	244
1	0.74	0.83	0.78	490
accuracy			0.69	734
macro avg	0.64	0.62	0.62	734
weighted avg	0.67	0.69	0.67	734
ROC-AUC: 0.6413390766142523				

Figure 8: Random forest model evaluation report

To further understand the behavior of the random forest model, two types of visualizations were generated: a confusion matrix and a feature importance chart. The confusion matrix provided a visual representation of classification performance across both classes. As shown in Figure 1, the model successfully identified a large proportion of “Won” proposals, while also capturing a meaningful number of “Lost” cases. This balance is essential for real-world deployment, where misclassification of low-potential proposals could lead to resource inefficiencies.

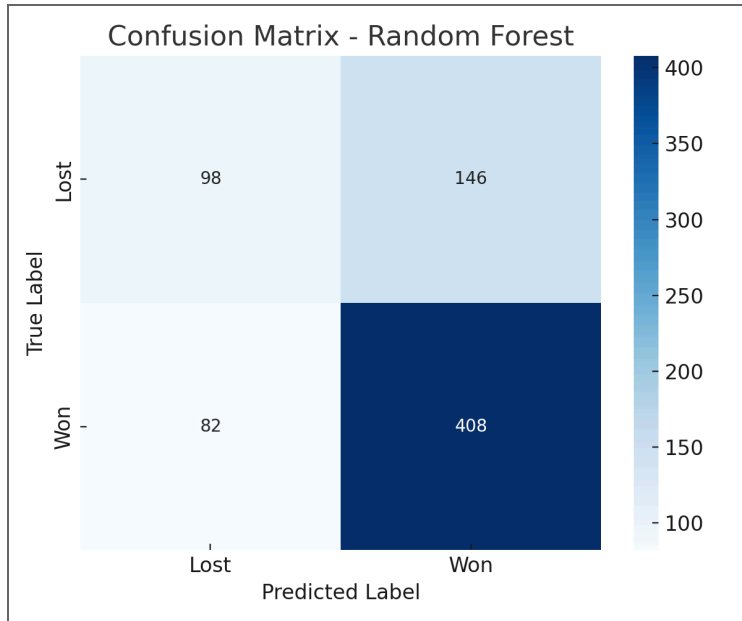


Figure 9: The random forest confusion matrix

The feature importance chart in Figure 10 below revealed the most influential features driving the model's predictions. Features such as contract award amount, region, commodity category, and organizational unit ranked highly, suggesting that both financial and contextual attributes played an important role in determining proposal success. This insight not only validates the choice of features but also offers actionable intelligence to business development teams seeking to optimize their bidding strategies.

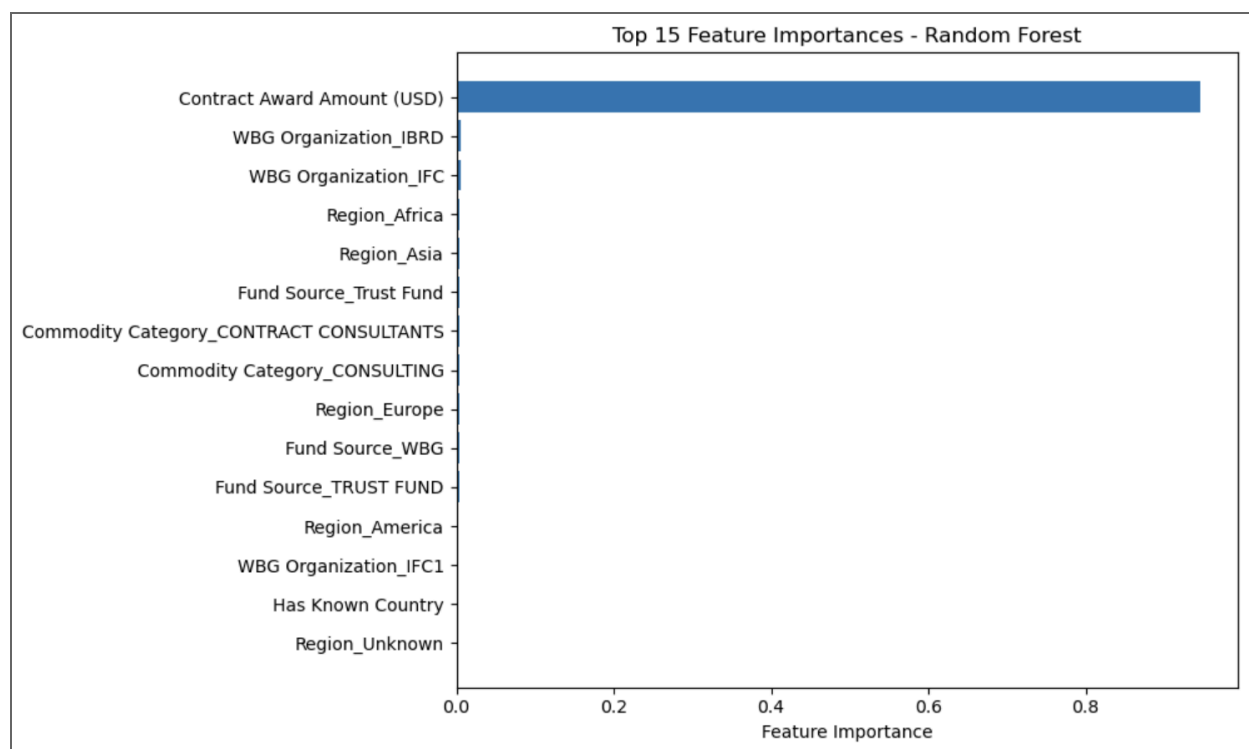


Figure 10: Top 15 important features

The evaluation results indicate that the random forest model is a robust and practical tool for improving opportunity identification in the consulting industry. Its superior performance over the baseline model justifies its selection as the primary model for deployment and further analysis.

6. Business Impact Analysis

Implementing a machine learning-based opportunity identification system presents significant business value for a consulting firm operating in a competitive, project-driven environment. By automating the evaluation of RFPs and estimating the probability of success for each proposal, the firm can shift from a reactive, intuition-driven business development approach to a more data-informed, proactive strategy. The random forest model developed and evaluated in this project offers a strong foundation for such transformation, demonstrating a reliable ability to differentiate between proposals likely to be awarded and those with lower success probability.

One of the most immediate benefits of the model is the enhancement of efficiency in the proposal scoping process. As previously estimated, business development

analysts at the firm spend approximately 12 hours per week searching for and evaluating RFPs. Across five practice areas and 20 business development staff, this represents 240 hours per week. By integrating the model into the RFP identification workflow, the firm can potentially reduce this effort by at least 70 percent, saving approximately 168 hours per week. An average hourly rate of \$90 per analyst translates to a weekly cost savings of \$15,120 and an annual savings of over \$750,000.

Beyond time and cost efficiency, the model enables more strategic resource allocation. The system guides BD teams to invest their effort in high-potential proposals by scoring and ranking opportunities based on historical success patterns. This reduces the likelihood of investing in bids poorly aligned with the firm's expertise or funding agency priorities. Over time, such prioritization could lead to higher win rates, more consistent revenue flow, and increased client trust.

Moreover, the model offers a competitive advantage in the fast-paced consulting landscape. Identifying relevant opportunities early - often days or weeks before manual scanning would have surfaced them, allows the firm to begin proposal development sooner, mobilize the most qualified personnel, and prepare stronger technical and financial submissions. This speed and responsiveness can distinguish between a successful and a missed opportunity.

The feature importance analysis adds an additional layer of strategic insight by highlighting the factors that most predict success. For example, if proposals in specific regions or commodity categories consistently rank higher, the firm might choose to develop deeper expertise or form partnerships in those areas. Similarly, organizational units that award contracts more frequently could become focal points for relationship-building and marketing efforts.

Significantly, the business impact of this model extends beyond individual bid decisions. It can become a knowledge asset integrated with internal data systems, continuously learning from new submissions and outcomes and serving as a feedback mechanism for refining business development strategies across the organization. When deployed effectively, this system represents not just a tool for automation but a lever for organizational learning and long-term growth.

7. Deployment Strategy

Deploying the machine learning solution for opportunity identification requires a structured approach that integrates seamlessly with the consulting firm's existing business development workflows. The goal is to ensure that model insights are delivered in a timely, accessible, and actionable manner for business development teams.

The deployment should begin with integrating the model into a user-facing dashboard or platform that aggregates RFPs and provides real-time scoring of opportunities. This dashboard can be developed using existing business intelligence tools such as Power BI, or through a custom web application, and should allow users to view ranked opportunities, explore key drivers behind the scores, and apply filters based on region, sector, or funding source.

From a technical standpoint, the model can be deployed using cloud services like AWS, Azure, or Google Cloud. Containerization with Docker and orchestration through CI/CD pipelines will enable scalable and automated updates. The scoring model can be hosted through a model-serving API, while RFPs can be ingested via scheduled scripts or connectors linked to procurement portals.

User adoption is critical, and the firm should prioritize training sessions to familiarize business development teams with the platform and model outputs. Feedback loops should be established so users can report the actual outcome of pursued proposals, enabling the model to improve continuously.

Governance measures must ensure responsible use of the model, including confidence thresholds for decision-making and human oversight for edge cases. With phased deployment, technical infrastructure, and organizational alignment in place, the model can be an integral part of the firm's business development strategy, improving efficiency and competitiveness in opportunity scoping.

8. Recommendations

While the development and deployment of the machine learning solution mark a significant advancement in automating opportunity identification, there remain several avenues for future improvement and expansion. These enhancements can further increase the accuracy, adaptability, and strategic value of the model over time.

First, the model can benefit from incorporating additional data sources beyond those available in the current dataset. Integrating internal firm-specific data such as past proposal narratives, technical approaches, team compositions, and client feedback could enrich the feature set and provide a more nuanced understanding of factors influencing success. Similarly, extending the scraping and aggregation pipeline to include real-time RFPs from a wider range of procurement platforms would ensure broader market coverage and greater opportunity visibility.

Technically, future model iterations could leverage advanced natural language processing (NLP) techniques to extract deeper insights from unstructured text in RFPs. While our analysis used structured data features such as contract amount and region, models like BERT or RoBERTa could analyze the semantic content of contract descriptions, enabling the model to detect thematic alignment between RFPs and the firm's core competencies. These techniques could be especially useful in identifying "hidden" opportunities that keyword-based methods might miss.

In terms of modeling, further experimentation with alternative algorithms such as gradient boosting machines (e.g., XGBoost or LightGBM) or neural networks could potentially yield improved performance.

Operationally, future versions of the system could support automated RFP alerts, bid/no-bid recommendations, and dynamic dashboards for tracking win rates, regional trends, and proposal performance over time. Establishing a feedback mechanism for continuously updating the model with new outcomes would ensure that the system remains current and aligned with evolving business dynamics.

9. Annex

9.1. Annex 1

[Python code](#) for the ML model development and evaluation

9.2. Annex 2

Part One of the AI/ML Business Use Case. Please see attached below:

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AI/ML Business Use Case

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1. Industry Overview: Management and Economics Consulting

The Management and Economics Consulting industry plays a key role in shaping economic policies, guiding financial strategies, and improving development outcomes across various sectors in the world. Firms in this industry provide advisory services to a wide range of clients, including governments, multilateral organizations, donor agencies, corporations, and financial institutions. These firms usually specialize in areas like financial services, human development, regulatory compliance, and public health. Consulting firms would then use their analytical approach and industry expertise to deliver (evidence-based) solutions that drive socio-economic progress.

Consulting operates under a highly competitive, project-based revenue model. Firms secure work primarily by responding to **Requests for Proposals (RFPs)** issued by clients seeking specialized expertise to address complex challenges. These RFPs outline the scope of work, budget considerations, evaluation criteria, and expected deliverables. As part of the business development (BD) process, firms must continuously scan various RFP portals, procurement websites, and client-specific platforms to identify suitable opportunities. Once an RFP is identified, firms invest significant effort in preparing compelling proposals, showcasing their qualifications, methodologies, and past performance to increase their chances of winning contracts. This pipeline of submitting proposals is resource-intensive and can often be inefficient.

To remain competitive and ahead, an AI-driven approach can significantly improve efficiency by automating RFP scanning, extracting key insights, and ranking opportunities based on historical win rates and strategic fit. The following sections outline the challenges associated with traditional BD processes, propose an ML-based solution, and detail the potential return on investment (ROI), implementation strategy, and ethical considerations.

2. Problem Identification

One major challenge for consulting firms is identifying relevant RFPs among the vast number published across multiple donor and government procurement portals. Beyond this, multiple other challenges exist, and includes:

- **Time-Intensive Manual Search:** Business development teams manually scan dozens of online platforms, procurement databases, and government websites daily to identify relevant opportunities. This manual activity is both time-consuming and prone to human oversight, hence difficult to consistently track all opportunities.
- **Missed Opportunities:** Due to the sheer volume of RFPs, many consulting firms fail to identify all relevant opportunities in time. Some opportunities remain hidden in less-known procurement portals, while others may be

overlooked due to inefficient keyword-based searches that do not capture all possible variations of relevant topics.

- **Inconsistent Relevance Matching:** Once RFPs are identified, BD teams must assess their alignment with the firm's capabilities, previous work, and working strategy. This assessment is often subjective and varies by team members, leading to inconsistent prioritization and poor resource allocation.
- **Delayed Response Time:** By the time an RFP is identified and prioritised, BD teams may have limited time to prepare a high-quality proposal. Tight deadlines reduce the ability to craft compelling narratives, conduct thorough research, and pull the right team members for the bid.
- **Competitive Disadvantage:** The manual approach to opportunity identification results in inefficiencies that put consulting firms at a competitive disadvantage. Firms that are slower in identifying and responding to opportunities may lose out to competitors who can act faster and submit more polished proposals.

3. ML-AI Solution Proposal

Automated Opportunity Identification & Ranking System

I have worked in economics consulting for 3+ years, and now have understood that addressing these challenges must require a more technology-driven approach. I am proposing a **Machine Learning (ML)-powered Opportunity Identification and Ranking System**. This AI-driven solution will automate the search, evaluation, and ranking of relevant RFPs, ensuring a faster and more accurate business development process. The key features would include:

1. **Web Scraping & Aggregation:** The system will continuously scan government portals, donor agency websites, and procurement platforms to collect RFPs in real-time. This will eliminate the need for manual searches, ensuring that all opportunities are captured promptly.
2. **Natural Language Processing (NLP) for RFP Analysis:** NLP techniques will be used to extract key details from RFP documents, including project scope, eligibility criteria, budget, deadlines, and thematic focus. The AI model will standardize and structure this data for easy comparison.
3. **Opportunity Scoring & Matching:** A machine learning algorithm will assess the RFPs based on multiple factors, such as alignment with historical win rates, firm expertise, strategic priorities, and resource availability. The system will generate an opportunity score to rank RFPs by relevance and potential success probability.
4. **Real-Time Alerts & Dashboards:** The BD teams will receive automatic notifications for high-relevance RFPs, ensuring prompt responses. A dashboard will provide an overview of key trends, successful bidding patterns, and market demand analysis.
5. **Adaptive Learning & Continuous Improvement:** The ML model will refine its predictions over time by incorporating feedback from successful and

unsuccessful proposals, enhancing accuracy in identifying high-potential opportunities.

Through this AI-driven automation, the solution will significantly reduce manual effort, improve decision-making, and enhance organisations' ability to secure competitive contracts. Implementing an automated system like this will streamline BD operations and also provide a significant competitive edge in the consulting industry.

4. ROI Analysis

Implementing this ML-driven Opportunity Identification and Ranking System will save substantial time and cost, and directly impact a firm's profitability and efficiency. For '**Company A**',

Time Savings Calculation:

- **Firm size:** 200 employees
- **Number of actively working practice areas:** 5
- **Average BD staff per practice area:** 4
- **Total BD staff across the firm:** $5 \times 4 = 20$
- **Current weekly hours spent per BD staff on opportunity scoping:** 12
- **Total weekly hours spent across all BD staff:** $20 \times 12 = 240$ hours
- **Projected reduction in manual effort with ML system:** 70%
- **New weekly hours spent:** $240 \times 0.3 = 72$ hours
- **Weekly hours saved:** $240 - 72 = 168$ hours

Financial Savings Calculation:

- **Average hourly rate of a BD Analyst:** \$90
- **Total weekly cost savings:** $168 \times \$90 = \$15,120$
- **Annual cost savings (assuming 50 work weeks):** $\$15,120 \times 50 =$
\$756,000

Additional Benefits:

- **Higher Proposal Success Rate:** By ensuring early identification of opportunities, BD teams can craft higher-quality proposals, increasing the win rate.
- **Increased Revenue Potential:** More successful bids lead to higher revenue generation for the firm.
- **Competitive Advantage:** Faster and more accurate opportunity identification allows the firm to stay ahead of competitors.

Potential Costs & Risks:

- **Implementation Costs:** AI model development, system integration, and user training.

- **Data Accuracy Issues:** Potential misclassification of opportunities requiring continuous model refinement.
- **Adoption Resistance:** BD teams may require change management strategies to fully embrace the new system.

By automating opportunity identification, **Company A** will gain substantial cost savings while improving efficiency and competitiveness.

5. Implementation Strategy

This process and development can be carried out in the following phased approach, safely assuming compatibility with firm infrastructure:

Phase 1: Needs Assessment & Data Collection (2 Months)

- Conduct stakeholder interviews to refine system requirements. Stakeholder includes team leads and partners in the various practice areas.
- Identify key RFP sources and procurement databases for integration. This will be led by the BD staff.
- Gather historical RFP data and successful bid records for model training.

Phase 2: System Development & Testing (4 Months)

- Develop the web scraping and NLP processing modules.
- Train the ML model on historical data to improve opportunity ranking accuracy.
- Conduct initial system testing with BD teams to gather feedback.

Phase 3: Deployment & Training (1 Month)

- Deploy the system across all BD practice areas.
- Conduct training sessions to ensure smooth adoption by BD teams.
- Monitor early performance and refine the model based on user feedback.

Phase 4: Optimization & Scaling (Monthly or Quarterly)

- Implement continuous learning for improved accuracy.
- Expand to additional procurement platforms and funding sources.
- Evaluate ROI and make necessary system enhancements.

Resources Needed:

- **Personnel:**
 - Data Scientists (2 members)
 - AI Engineers (2 members)
 - Business Intelligence Analysts (1 member)
 - BD Specialists (10 members, 2 per practice area)
 - IT Support (IT team)
 - Change Management Experts (2–3 members)

- **Technology Stack:**
 - Cloud computing (AWS, Azure, or Google Cloud)
 - NLP frameworks (SpaCy, BERT)
 - Web Scraping tools
 - IDE for programming (Python)
 - Data visualization tools (Power BI)
 - CI/CD pipelines.

6. Ethical Considerations

Bias and Fairness:

ML models may introduce biases if trained on unbalanced datasets. The system may underrepresent some opportunities if past successful proposals disproportionately favor certain topics or funders. Regular audits and human oversight should be in place to mitigate such risks.

Transparency and Explainability:

For the BD team, a lack of transparency could lead to distrust in AI-driven recommendations. The system should provide interpretable justifications for its rankings and allow manual adjustments when necessary.

Data Privacy and Security:

Procurement and bidding data are often sensitive business assets. The system must implement strong cybersecurity measures, including access controls and compliance with company data use regulations to prevent unauthorized access or data breaches.

Job Displacement Concerns:

While automation enhances efficiency, it should be positioned as a tool to support rather than replace human expertise. Therefore BD teams should be trained to work alongside AI.

Regulatory Compliance:

AI-driven solutions must adhere to ethical bidding practices and procurement regulations to prevent unfair competitive advantages. Compliance measures should be regularly reviewed to align with industry and legal standards.

By consciously addressing these ethical considerations, consulting firms can maximize the benefits of AI while ensuring fairness, security, and regulatory adherence in their business development processes.